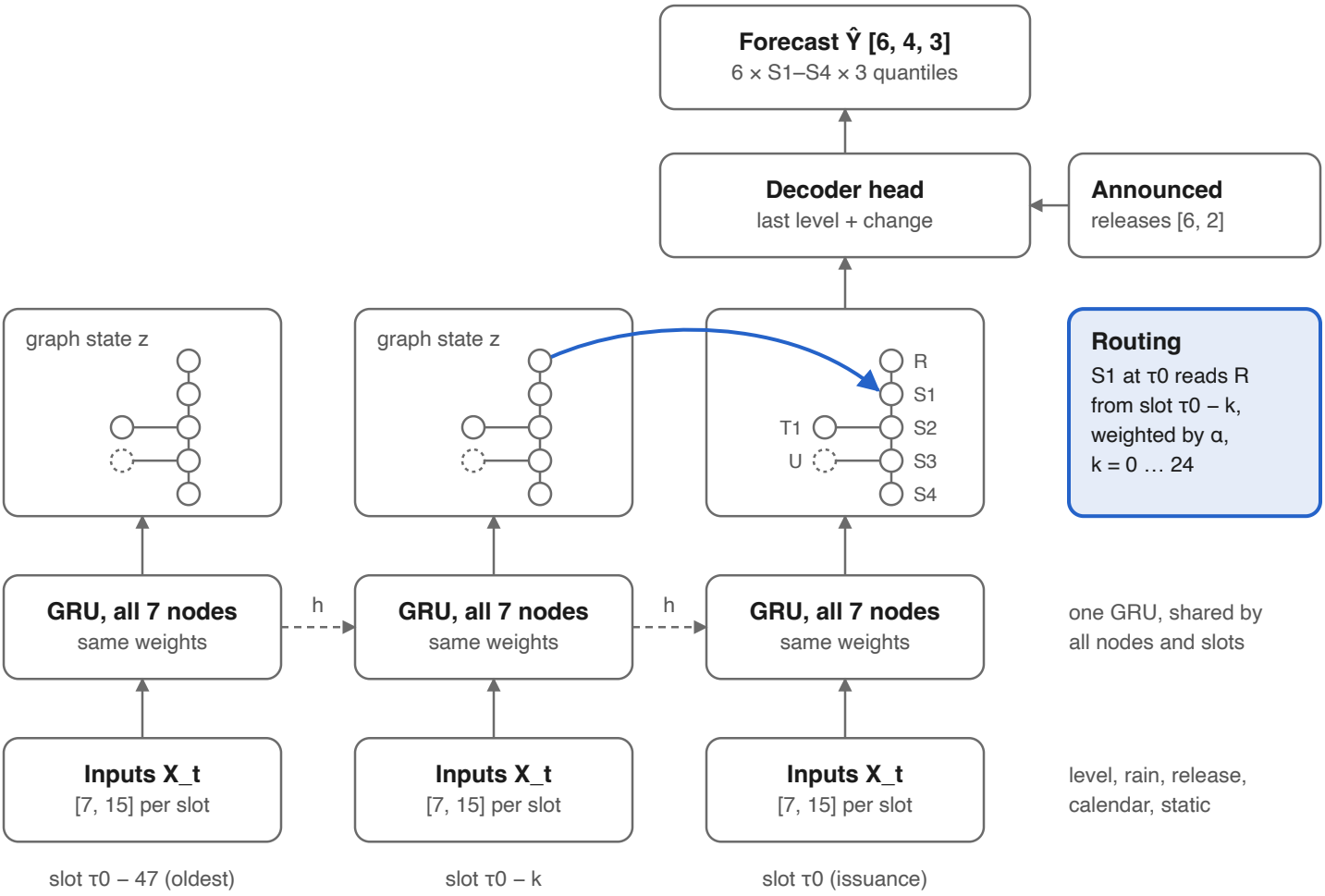Unrolled in time: shared GRU per node, lagged routing, one head at τ0